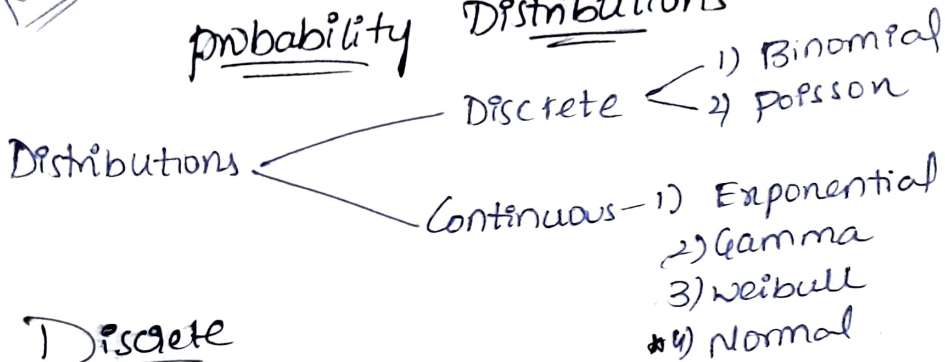


## Module - 4

# probability Distributions



Discrete

(1) Binomial  $0, 1, 2, \dots, n$  (particular case).

(2) Poisson  $t = 0, 1, 2, \dots, \infty$

$$B \cdot D \subseteq P \cdot D$$

## Notations

$n, p, q, x$   
 $\downarrow \quad \downarrow \quad \downarrow$   
 succ Fail not a thing  
 we are

\* Binomial distribution  $p(x) = n_c p^n q^{n-x}, x=0, 1, 2, \dots, n.$

\* mean of the Binomial distribution =  $np$ .

$$\boxed{\text{mean} = np}$$

$$\text{Variance} = npq$$

probab mean of B.D is 5, S.D is 2. Determine deviation

$\sum \frac{f_i}{2} \times \text{mean} = np = 5 \quad \text{--- (1)}$

$$\text{Variance} = \sqrt{npq} = 2$$

$$p(x) = n C_x p^x q^{n-x}, \quad x=0, 1, 2, \dots, n$$

$$\sqrt{npq} = 2 \Rightarrow npq = 4 \quad \text{--- (2)}$$

$$\frac{(2)}{(1)} \Rightarrow \frac{npq}{np} = \frac{4}{5} \Rightarrow \boxed{q = \frac{4}{5}} \Rightarrow p = 1 - \frac{4}{5} = \frac{1}{5} \Rightarrow \boxed{p = \frac{1}{5}}$$

$$\therefore np = 5 \Rightarrow n\left(\frac{1}{5}\right) = 5 \Rightarrow \boxed{n = 25}$$

$$\therefore p(x) = {}^{25}C_x \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{25-x} \Rightarrow \text{Required distribn}$$

\* mean =  $np = \frac{4}{3}$  ,  $p(x \geq 1)$

$$np = 4, npq = \frac{4}{3} \Rightarrow \boxed{q = \frac{1}{3}}, \boxed{p = \frac{2}{3}} \quad n\left(\frac{2}{3}\right) = 4$$

$$p(x) = {}^6C_x \left(\frac{2}{3}\right)^x \left(\frac{1}{3}\right)^{6-x}, x = 0, 1, 2, \dots, 6.$$

$$\begin{aligned} p(x \geq 1) &= 1 - p(x < 1) \\ &= 1 - p(x = 0) \\ &= 1 - {}^6C_0 \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^{6-0} \\ &= 1 - 1 \left(\frac{1}{729}\right) \\ &= \frac{728}{729} // \\ &= 0.998 // \end{aligned}$$

\* An irregular 6 faced die is such that probability that it gives 3 even numbers in 5 throws is twice the probability that it gives 2 even nos in 5 throws. How many sets of exactly 4 trials can be expected to give no even nos out of 2500 sets.

let  $x$  denotes no. of even numbers obtained in 5 <sup>throws/trials</sup> trials

$$\boxed{p(x=3) = 2 * p(x=2)}$$

$$\boxed{n_{cr} = \frac{n!}{(n-r)! r!}}$$

$${}^5C_3 p^3 q^{5-3} = 2 \times {}^5C_2 p^2 q^{5-2}$$

$$\frac{5 \times 4 \times 3}{3 \times 2 \times 1} p^3 q^2 = 2 \times \frac{5 \times 4}{2 \times 1} p^2 q^3$$

$$\cancel{10} p^3 q^2 = 2 \times \cancel{10} p^2 q^3$$

$$p = 2q$$

$$p = 2(1-p) \Rightarrow p = 2 - 2p \Rightarrow 3p = 2 \Rightarrow \boxed{p = \frac{2}{3}}$$

$$\boxed{q = \frac{1}{3}}$$

$$\therefore p(x=0) = {}^5C_0 \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^5 \Rightarrow \frac{1}{243} // = 0.00411 //$$

$$\text{out of 2500 sets} \Rightarrow 2500 \times \frac{0.00411}{p(x=0)} = 10.275 //$$

\* Poisson distribution:-

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$x = 0, 1, 2, \dots, \infty$

Here,

$$\text{mean} = \lambda = np = \text{Variance}$$

- Ex: No. of defective articles in packet of 100.  
 No. of Printing errors at each page of book.  
 No. of road accidents - - -

\* The number of monthly breakdowns of a computer is a random variable having P.D with mean equal to 1.8. Find probability that this computer will fail for month.

- without breakdown,
- with only one breakdown,
- with atleast one breakdown,

Sol:  $P(x) = \frac{e^{-\lambda} \lambda^x}{x!}$  ;  $p(x) = \frac{e^{-1.8} (1.8)^x}{x!}$  ,  $x = 0, 1, 2, \dots, \infty$

$x$  = no. of breakdown,

a)  $p(x=0) = \frac{e^{-1.8} (1.8)^0}{0!} = e^{-1.8} = 0.165 //$

b)  $p(x=1) = \frac{e^{-1.8} (1.8)^1}{1!} = 0.165 \times 1.8 = 0.297 //$

c)  $p(x \geq 1) = 1 - p(x \leq 0) = 1 - p(x=0) = 1 - 0.165 = 0.835 //$

\* If the probability of defective fuse from manufactory unit is 2%. in box of 200 fuses, find the probability that a) Exactly 4 fuses are defective.  
 b) more than 3 fuses are defective.

Sol:  $p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$  ,  $x = 0, 1, 2, \dots, \infty$

$p = 2\% = \frac{2}{100} = 0.02$  ,  $n = 200$ .

$\lambda = np = 200 \times 0.02 = 4 //$

$p(x) = \frac{e^{-4} (4)^x}{x!}$  ,  $x = 0, 1, 2, \dots, \infty$

$$a) p(x=4) = \frac{e^{-4}(4)^4}{4!} = \frac{0.018 \times 256}{24} = \frac{4.608}{24} = 0.192 //$$

$$b) p(x > 3) = 1 - p(x \leq 3) = 1 - [p(x=0) + p(x=1) + p(x=2) + p(x=3)]$$

$$= 1 - \frac{e^{-4}(4)^0}{0!} + \frac{e^{-4}(4)^1}{1!} + \frac{e^{-4}(4)^2}{2!} + \frac{e^{-4}(4)^3}{3!}$$

$$= 1 - 0.018 \times \left[ 1 + 4 + \frac{16}{2} + \frac{64}{6} \right]$$

$$= 1 - 0.018 \times [1 + 4 + 8 + 10.66]$$

$$= 1 - 0.018 [23.66]$$

$$= 1 - 0.42588$$

Fitting of Binomial = 0.56842 //

\* Theoretical distribution (Fitting Binomial (poisson distribn))

~~Problem~~  $N \times p(x=x)$

\* Fit binomial distribn for following data -  
problem

| $x$ | $f$ | $fx$ | T. freq |
|-----|-----|------|---------|
| 0   | 5   | 0    | 4       |
| 1   | 18  | 18   | 15      |
| 2   | 28  | 56   | 25      |
| 3   | 12  | 36   | 22      |
| 4   | 7   | 28   | 11      |
| 5   | 6   | 30   | 3       |
| 6   | 4   | 24   | 0       |

$\Sigma f = 80$   $\Sigma fx = 192$

$n=6$

Theoretical to Binomial

$$N \times p(x) \Rightarrow 80 \times {}^6C_x (0.4)^x (0.6)^{6-x}; x=0, 1, 2, \dots, 6$$

| $x$    | 0    | 1     | 2     | 3     | 4     | 5    | 6 |
|--------|------|-------|-------|-------|-------|------|---|
| $P(x)$ | 8.73 | 14.93 | 24.88 | 22.12 | 11.06 | 2.95 |   |

Poisson to Binomial

$$p(x) = {}^nC_x p^x q^{n-x}, x=0, 1, \dots, n$$

$$\lambda = np \Rightarrow \bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{192}{80}$$

$$np = \frac{192}{80} = 2.4$$

$$6p = \frac{192}{80}$$

$p=0.4$

$$q = 1 - p = 1 - 0.4 = 0.6 //$$

$\bar{x} = 2.4$  =  $\lambda$

$$\lambda = np = 2.4$$

# Binomial to Poisson

$$N \times p(x) = 80 \times \frac{e^{-2.4} (2.4)^x}{x!}$$

$$80 \times e^{-2.4} \Rightarrow 7.25$$

Enter

| x    | 0     | 1     | 2     | 3      | 4      | 5     | 6     |
|------|-------|-------|-------|--------|--------|-------|-------|
| P(x) | 7.257 | 17.41 | 20.88 | 16.704 | 10.022 | 4.810 | 1.924 |

Fitting of Poisson Distribution

| Deaths    | 0   | 1  | 2  | 3 | 4 |
|-----------|-----|----|----|---|---|
| frequency | 122 | 60 | 15 | 2 | 1 |

| x | f   | fx | Theo. freq. |
|---|-----|----|-------------|
| 0 | 122 | 0  | 121         |
| 1 | 60  | 60 | 61          |
| 2 | 15  | 30 | 15          |
| 3 | 2   | 6  | 3           |
| 4 | 1   | 4  | 0           |

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{100}{200} = 0.5$$

$$\bar{x} = 0.5$$

$$\lambda = \bar{x}$$

here

To dist given by  $\Rightarrow N \times P(X=x) \rightarrow 200 \times \frac{e^{-\lambda} \lambda^x}{x!}$

$$\Rightarrow 200 \times \frac{e^{-0.5} (0.5)^x}{x!} \Rightarrow$$

$$\Rightarrow$$

| x | 0   | 1  | 2  | 3 | 4 |
|---|-----|----|----|---|---|
| f | 121 | 61 | 15 | 3 | 0 |



(1) Exponential distribution

$$f(x|\lambda) = \lambda e^{-\lambda x}; x > 0, \lambda > 0$$

$$f(x) = \lambda e^{-\lambda x}; x > 0, \lambda > 0 \quad \text{also known as negative exponential distn}$$

\* mean =  $1/\lambda$  ;

\*  $r^{\text{th}}$  moment about origin is  $\frac{r!}{\lambda^r}$

\* Variance =  $\frac{1}{\lambda^2}$  ;

\* mileage which car owners get with certain brand of radial tires is RV having an exponential distribution with mean 40,000 km. Find probabilities that one of these tires will last  
 (i) at least 20,000 km (ii) At most 30,000 km

Solr  $f(x) = \lambda e^{-\lambda x}, \lambda > 0, x > 0$

let  $x$  be RV defined as mileage of car.

mean  $\Rightarrow \frac{1}{\lambda} = 40,000$

$$\lambda = \frac{1}{40,000}$$

$$f(x) = \frac{1}{40,000} e^{-x/40,000}, x > 0$$

$$P(x \geq x) = \int_x^{\infty} f(x) dx$$

(i) At least 20,000 km

$$P(x \geq 20,000) = \int_{20,000}^{\infty} \frac{1}{40,000} e^{-x/40,000} dx$$

$$= \frac{1}{40,000} \left[ \frac{e^{-x/40,000}}{-1/40,000} \right]_{20,000}^{\infty}$$

$$\Rightarrow \frac{1}{40,000} \left[ \frac{-1}{40,000} \left( e^{\frac{x}{40,000}} \right)_{12}^{\infty} \right]$$

$$\Rightarrow \left[ e^{-x/40,000} \right]_{12}^{\infty} = \frac{1}{40,000} \left[ -e^{-\infty} + e^{-0.5} \right]$$

$$= [0 + 0.6065]$$

$$= 0.6065 //$$

$$\begin{aligned}
 \text{iii) } P(X \leq 30,000) &= \frac{1}{40,000} \int_0^{30,000} (e^{-x/40,000}) dx \\
 &= \frac{1}{40,000} \left[ \frac{e^{-x/40,000}}{-1/40,000} \right]_0^{30,000}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{40,000} [e^{-30,000/40,000} + e^0] \\
 &= [-e^{-3/4} + 1]
 \end{aligned}$$

$$\begin{aligned}
 &= [1 - e^{-3/4}] \\
 &= 1 - 0.472
 \end{aligned}$$

$$= 0.5287$$

\* The time (in hours) required to repair a machine is exponentially distributed with parameter  $\lambda = 1/2$ .

a) What is the probability that repair time exceeds 2h?

b) What is the conditional probability that a repair takes at least 10h given that its duration exceeds 4h?

$$b) p(x \geq 10 | x > 9) = \frac{p(x \geq 10, x > 9)}{p(x > 9)}$$

$$p(x \geq 10) = 1 - \frac{1}{2} \int_0^{10} e^{-x/2} dx \Rightarrow 1 - \frac{1}{2} \left[ \frac{e^{-x/2}}{-1/2} \right]_0^{10}$$

$$\Rightarrow 1 - \frac{1}{2} \left[ \frac{e^{-5} - 1}{-1/2} \right] \Rightarrow 1 - \cancel{\frac{1}{2}} \left[ \frac{e^{-5} - 1}{\cancel{-1/2}} \right] \Rightarrow \cancel{1} + e^{-5} + \cancel{1}$$

$$= 6.73 \times 10^{-3}$$

$$= 0.00673 //$$

$$p(x > 9) = 1 - \frac{1}{2} \int_0^9 e^{-x/2} dx \Rightarrow 1 - \frac{1}{2} \left[ \frac{e^{-x/2}}{-1/2} \right]_0^9$$

$$\Rightarrow 1 - \frac{1}{2} \left[ \frac{e^{-4.5} - 1}{-1/2} \right] \Rightarrow \cancel{1} + e^{-4.5} + \cancel{1} \Rightarrow e^{-4.5} = 0.01118$$

$$\Rightarrow \frac{p(x \geq 10 | x > 9)}{p(x > 9)} = \frac{0.00673}{0.01118} = 0.6058 //$$

$$(a) p(x > 2) = 1 - \frac{1}{2} \int_0^2 e^{-x/2} dx \Rightarrow 1 - \frac{1}{2} \left[ \frac{e^{-x/2}}{-1/2} \right]_0^2$$

$$\Rightarrow 1 - \frac{1}{2} \left( \frac{e^{-1} - 1}{-1/2} \right) \Rightarrow 1 - \frac{1}{2} \left( \frac{e^{-1} - 1}{\cancel{-1/2}} \right) \Rightarrow 1 - \cancel{\frac{1}{2}} (e^{-1} - 1) (\cancel{-2})$$

$$\Rightarrow \cancel{1} + e^{-1} + \cancel{1} \Rightarrow e^{-1} = 0.367 //$$



## (g) Normal Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$-\infty < x < \infty, -\infty < \mu < \infty, \sigma > 0$$

$\mu$  = mean

$\sigma$  = standard deviation

⇒ Standard normal random variable

$$Z = \frac{x-\mu}{\sigma}$$

$$Z \rightarrow \infty \text{ as } x \rightarrow \infty$$

Sol

The marks obtained by no. of students in certain Subject are approximately normally distributed with mean 65, S.D 5. If 3 students are selected at random from this group, what is prob-ty atleast 1 of them would have scored above 75?

Sol:  $\mu = 65; \sigma = 5$

Let  $x$  be the R.V as marks.

$$P(x > 75) = P(75 < x < \infty)$$

$$Z = \frac{x-65}{5}$$

$$\Rightarrow P\left(\frac{75-65}{5} < \frac{x-65}{5} < \infty\right)$$

$$= P(2 < Z < \infty)$$

$$= P(0 < Z < \infty) - P(0 < Z < 2)$$

$$= 0.5 - 0.4772$$

$$P(x > 75) = 0.0228 //$$

$$P = 0.0228 //$$

$$P(X \geq 1) = {}^nC_x p^x q^{n-x}, x=0, 1, \dots, n$$

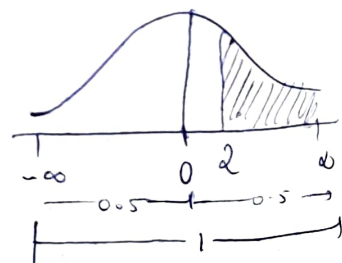
$$P(X \geq 1) = 1 - P(X < 1)$$

$$= 1 - P(X = 0)$$

$$= 1 - {}^3C_0 (0.0228)^0 (0.9772)^{3-0}$$

$$= 1 - 0.9331$$

$$= 0.0669 //$$



$$P = 0.0228$$

$$Q = 1 - 0.0228$$

$$= 0.9772$$

\* In Engineering exam, student is considered to be failed, secured second class, first class and distinction, a/c to he scored less than 45%, b/w 45% & 60%, b/w 60% & 75% and above 75% respectively. In particular year 10% of students failed in examination, 5% of students got distinction. Find % of students who have got first class and second class. (Assume normal distribution of marks). Given:-

Sol<sup>n</sup>  $F \Rightarrow < 45$   
 $II \Rightarrow 45 \text{ to } 60$   
 $III \Rightarrow 60 \text{ to } 75$   
 $D \Rightarrow > 75$

$$P(X < 45) = 10\%$$

$$P(X > 75) = 5\%$$

let  $X$  be percentage of students.

$$\text{To find } P(45 < X < 60) = ?$$

$$P(60 < X < 75) = ?$$

from given

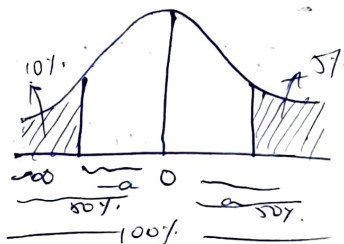
$$P(-\infty < X < 45) = P\left(-\infty < \frac{X - \mu}{\sigma} < \frac{45 - \mu}{\sigma}\right)$$

$$= P\left(-\infty < Z < \frac{45 - \mu}{\sigma}\right)$$

$$= 0.10 //$$

$$P(75 < X < \infty) = P\left(\frac{75 - \mu}{\sigma} < Z < \infty\right)$$

$$= 0.05 //$$



$$P\left(0 < Z < \frac{\mu - 45}{\sigma}\right) = 0.40$$

$$P\left(0 < Z < \frac{75 - \mu}{\sigma}\right) = 0.45$$

$$\frac{\mu - 45}{\sigma} = 0.40$$

$$\frac{\mu - 45}{\sigma} = Z = 1.28 //$$

(from clerk's table)

$$\frac{75 - \mu}{\sigma} = Z = 1.64 \text{ or } 1.65$$

$$\frac{75 - \mu}{\sigma} = 1.64 //$$

$$\boxed{\mu + 1.64\sigma = 75} \quad \text{--- (2)}$$

$$\boxed{\mu - 1.28\sigma = 45} \quad \text{--- (1)}$$

Solve (1) & (2)  $\Rightarrow \mu = 58.15$   
 $\sigma = 10.24 //$

$$P(45 < X < 60) \Rightarrow P\left(\frac{45 - \mu}{\sigma} < Z < \frac{60 - \mu}{\sigma}\right)$$

$$\Rightarrow P\left(\frac{45 - 58.15}{10.27} < Z < \frac{60 - 58.15}{10.27}\right)$$

$$\Rightarrow P(-1.28 < Z < 0.18)$$

$$\Rightarrow P(0 < Z < 1.28) + P(0 < Z < 0.18)$$

$$\Rightarrow 0.3997 + 0.0714$$

$$\Rightarrow 0.4711 //$$

$\therefore 47.11\%$  of students got Second class.

$$//y P\left(\frac{60 - 58.15}{10.27} < Z < \frac{75 - 58.15}{10.27}\right)$$

$$\Rightarrow P(0.18 < Z < 1.64)$$

$$\Rightarrow P(0 < Z < 0.18) + P(0 < Z < 1.64)$$

$$\Rightarrow 0.0714 + 0.4495$$

$$\Rightarrow P(0 < Z < 1.64) - P(0 < Z < 0.18)$$

$$\Rightarrow 0.4495 - 0.0714$$

$$= 0.3781$$

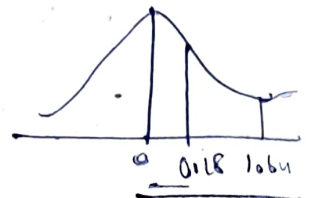
$\therefore 37.81\%$  of students got first class.

$$\% \text{ of students who got first class} = 100 - (47 + 5 + 10)$$

$$= 100 - (47 + 15)$$

$$= 100 - 62$$

$$= 38\%$$



### 3) Weibull Distribution

$$f(x, \alpha, \beta) \text{ or } f(x) = \begin{cases} \alpha \beta x^{\beta-1} e^{-\alpha x^\beta} & ; x > 0 \\ 0 & \text{otherwise} \end{cases}$$

• where  $\alpha > 0, \beta > 0$

•  $X \sim \text{Gamma}(\alpha, \beta)$

at least  $\geq 8$   
at most  $\leq 8$

Problem - The length of life  $x$ , in hours, of an item in a machine shop has a Weibull distribution with  $\alpha = 0.01$  and  $\beta = 2$ . What is the probability that it fails before eight hours of usage?

Given,  $\alpha = 0.01, \beta = 2$ .

let  $x$  = usage of hours.

$$P(X < 8) = ?$$

$$P(X < 8) = \int_0^8 f(x) dx$$

$$f(x) = (0.01)^2 x^{2-1} e^{-(0.01)x^2}, x > 0$$

$$= (0.01) 2x^1 e^{-(0.01)x^2}, x > 0$$

$$P(X < 8) = \int_0^8 (0.01) 2x e^{-(0.01)x^2} dx$$

$$= \int_0^{0.64} e^{-t} dt$$

$$= \left[ -e^{-t} \right]_0^{0.64}$$

$$= 1 - e^{-0.64}$$

$$= 0.472$$

let  $0.01 x^2 = t$   
 $0.01 2x dx = dt$

|     |   |      |
|-----|---|------|
| $x$ | 0 | 8    |
| $t$ | 0 | 0.64 |

## \* Gamma distribution (Erlang's) :-

$$f(x) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-\frac{1}{\beta}x}, & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

• where  $x > 0, \beta > 0$ .

•  $X \sim \text{Gamma}(\alpha, \beta)$ .

\* Exponential  $\sim \text{Gamma}(\alpha=1, \beta)$ .

$$f(x) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta},$$

$$\lambda = \frac{1}{\beta}$$

$$\alpha=1 \Rightarrow f(x) = \frac{1}{\beta} e^{-x/\beta}$$

$$\text{Sub } \frac{1}{\beta} = \lambda$$

$$\boxed{f(x) = \lambda e^{-\lambda x}} \Rightarrow \text{exponential.}$$

for  $k=3$

(For 10m)

## \* Problem 1

In Certain city, the daily consumption of electric power in millions of kilowatt-hours can be treated as RV having an Erlang distribution with parameters  $\lambda = \frac{1}{2}, k=3$ . If the power plant of this city has daily capacity of 12 millions k.w hours. What is the probability that this power supply will be inadequate on any given day.

Sol<sup>n</sup> let  $x$  be daily consumption of electric power.

Given,  $\lambda = \frac{1}{2}, k=3$

$$\lambda = \frac{1}{\beta} \Rightarrow \lambda = \frac{1}{2} = \frac{1}{\beta} \Rightarrow \boxed{\beta=2}, \boxed{\alpha=3}$$

$$P(X > 12) = \int_{12}^{\infty} f(x) dx.$$

$$f(x) = \frac{1}{2^3 \sqrt{3}} x^{3-1} e^{-\frac{x}{2}}, \quad x > 0$$

$$= \frac{1}{2^3 \sqrt{3}} x^2 e^{-x/2}, \quad x > 0$$



$$P(x > 12) = \int \frac{1}{16} x^2 e^{-x/2} dx$$

$$= \frac{1}{16} \left[ \frac{x^2 e^{-x/2}}{(-1/2)} - \frac{2x e^{-x/2}}{(-1/2)(-1/2)} + \frac{2e^{-x/2}}{(-1/2)(-1/2)(-1/2)} \right]$$

Integration by parts:

$$d(uv) = u'v + u v' - u''v - u'''v - \dots$$

Let  $u = x^2$ ,  $v = e^{-x/2}$

$$e^{-x/2} dx = dv$$

$$\frac{e^{-x/2}}{-1/2} = v$$

Integration by parts formula:

$$\int u'v dx = uv - \int u''v dx$$

$$\Rightarrow \frac{1}{16} \left[ \frac{x^2 e^{-x/2}}{(-1/2)} - \frac{2x e^{-x/2}}{(-1/2)(-1/2)} + \frac{2e^{-x/2}}{(-1/2)(-1/2)(-1/2)} \right]$$

$$\Rightarrow \frac{1}{16} \left[ \left( \frac{x^2 e^{-x/2}}{-1/2} \right) - \left( \frac{2x e^{-x/2}}{1/4} \right) - \left( \frac{2e^{-x/2}}{-1/8} \right) \right]$$

$$\Rightarrow \frac{1}{16} \left( \frac{\infty^2(0)}{-1/2} - \frac{1uv(e^{-6})}{-1/2} \right) \dots$$

$$\Rightarrow \frac{1}{16} \left[ e^{-6} (288 + 96 + 16) \right]$$

$$\Rightarrow 95 e^{-6}$$

$$\Rightarrow 0.0625 //$$